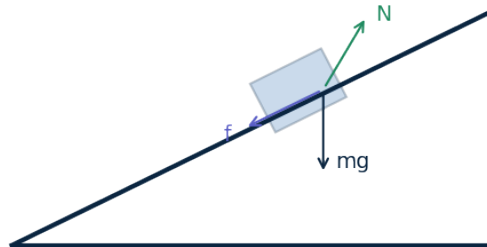


Laws of Motion

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Newton's First Law: a body remains at rest or in uniform motion unless acted upon by a net external force (law of inertia).
- Newton's Second Law: $F(\text{net}) = dp/dt = ma$, where $p = mv$ is momentum. Valid in an inertial frame.
- Newton's Third Law: for every action there is an equal and opposite reaction, acting on different bodies.
- Friction: static friction opposes the start of relative sliding ($f_s \leq \mu_s N$); kinetic friction acts during sliding ($f_k = \mu_k N$). Generally $\mu_k < \mu_s$.
- For a block on a rough incline at angle θ , it starts sliding when $\tan(\theta) \geq \mu_s$ (angle of repose).
- In a connected system (e.g., Atwood machine, blocks on a string over a pulley), draw a separate free-body diagram for each body and apply $F = ma$ to each.
- Pseudo force $= -m a(\text{frame})$ is used only in a non-inertial (accelerating) reference frame, acting opposite to the frame's acceleration.



Free-body diagram of a block on a rough inclined plane, showing mg , Normal force N , and friction f .

2. Formula Sheet

Newton's second law	$F = ma = dp/dt$
Static friction (max)	$f_s(\text{max}) = \mu_s N$
Kinetic friction	$f_k = \mu_k N$
Angle of repose	$\tan(\theta) = \mu_s$
Acceleration on smooth incline	$a = g \sin(\theta)$
Atwood machine acceleration	$a = (m_1 - m_2) g / (m_1 + m_2)$
Tension in Atwood machine	$T = 2 m_1 m_2 g / (m_1 + m_2)$
Apparent weight in a lift (acc. a up)	$N = m(g + a)$

3. Practice Problems (22 questions)

1. A block of mass 5 kg is pulled by a horizontal force of 20 N on a frictionless surface. Find its acceleration.
2. A force of 10 N acts on a body of mass 2 kg initially at rest. Find the velocity of the body after 4 s.
3. Two masses 4 kg and 6 kg are connected by a string over a frictionless pulley (Atwood machine). Find the acceleration of the system and the tension in the string ($g = 10 \text{ m/s}^2$).
4. A block of mass 10 kg rests on a rough horizontal surface with coefficient of static friction 0.4. Find the minimum horizontal force needed to move the block.
5. A block slides down a frictionless incline of angle 30 degrees. Find its acceleration.
6. A body of mass 2 kg is placed on a rough incline of angle 45 degrees with $\mu_k = 0.5$. Find its acceleration down the incline.
7. A man of mass 60 kg stands on a weighing machine inside a lift accelerating upward at 2 m/s^2 . Find the reading of the weighing machine.
8. Find the coefficient of friction if a block just begins to slide on an incline when the angle is 37 degrees.
9. A 1000 kg car experiences a resistive force of 500 N while moving at constant velocity. What driving force must the engine provide to maintain this constant velocity?
10. Two blocks of masses 3 kg and 2 kg are in contact on a frictionless floor. A force of 10 N is applied on the 3 kg block. Find the acceleration of the system and the contact force between the blocks.
11. A monkey of mass 20 kg climbs a rope that can withstand a tension of 250 N. Find the maximum acceleration with which the monkey can climb up.
12. State Newton's third law and explain why it does not lead to zero net force in a system of two interacting bodies.
13. A block of mass 4 kg is at rest on a rough surface ($\mu_s = 0.3$). Find the frictional force acting when a horizontal force of 10 N is applied.
14. A car of mass 1200 kg moving at 20 m/s is brought to rest in 4 s by applying brakes. Find the average braking force.

15. A ball of mass 0.2 kg hits a wall with a speed of 10 m/s and rebounds with the same speed. If the contact time is 0.01 s, find the average force exerted by the wall on the ball.
16. Explain why it is easier to pull a lawn roller than to push it, using the concept of the normal force.
17. A lift is descending with acceleration 3 m/s^2 . Find the apparent weight of a 50 kg person inside ($g = 10 \text{ m/s}^2$).
18. Two blocks of mass 2 kg and 3 kg connected by a light string are pulled by a horizontal force of 25 N on a smooth surface. Find the tension in the string.
19. A body of weight 200 N is pulled along a horizontal surface by a force making an angle of 60 degrees with the horizontal. If $\mu_k = 0.3$, find the frictional force (take the normal force reduced by the vertical component of the applied force, with applied force magnitude 100 N).
20. Why does a passenger standing in a bus lurch forward when the bus suddenly stops? Explain using Newton's first law.
21. A block of mass 5 kg is held stationary on a smooth incline of 30 degrees by a horizontal force F. Find F.
22. Derive the expression for the acceleration of two blocks connected over a pulley (Atwood machine) using free-body diagrams.

Answer Key

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|---|---|
| 1. $a = 4 \text{ m/s}^2$ | 2. $v = 20 \text{ m/s}$ |
| 3. $a = 2 \text{ m/s}^2$, $T = 48 \text{ N}$ | 4. $f = 40 \text{ N}$ |
| 5. $a = 5 \text{ m/s}^2$ | 6. $a = 3.54 \text{ m/s}^2$ (approx) |
| 7. Reading = 720 N (mass equivalent 72 kg) | 8. $\mu_s = 0.75$ |
| 9. $F = 500 \text{ N}$ | 10. $a = 2 \text{ m/s}^2$, contact force = 4 N |
| 11. $a = 2.5 \text{ m/s}^2$ | 12. Conceptual - action-reaction act on different bodies, so they don't cancel |
| 13. $f = 10 \text{ N}$ (static, since $10 \text{ N} < f_{s \text{ max}} = 11.76 \text{ N}$, block stays at rest) | 14. $F = 6000 \text{ N}$ |
| 15. $F = 400 \text{ N}$ | 16. Conceptual - pushing increases normal force, pulling decreases it, reducing friction when pulling |
| 17. Apparent weight = 350 N | 18. $T = 15 \text{ N}$ |
| 19. f approx 21 N (using reduced normal force) | 20. Conceptual - body tends to continue in motion (inertia) while bus decelerates |
| 21. $F = 5g \tan(30)$ approx 28.3 N | 22. Conceptual derivation - see notes above |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.